

FATE Monitoring: DOC Data

Spring & Summer 2025

2025-09-29

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##Setup - Change things here & write any notes

```
#identify section
cat("Setup Information")
```

Setup Information

```
##### Run information - PLEASE CHANGE
Date_Run = "9/19/25" #Date that instrument was run
Run_by = "Stephanie J. Wilson" #Instrument user
Script_run_by = "Stephanie J. Wilson" #Code user
run_notes = "TN duplicates were out of range, accepting because checks and other QA all fine." #any
samples <- c("FATE") #whatever identifies your samples within the same names
samples_pattern <- paste(samples, collapse = "|")
#samples_pattern <- "GCW" #use this instead of the line above if you have only one site code
chks_name = "Chk_Std_1CN" #what did you name your check standards?

##### File Names - PLEASE CHANGE
#file path and name for raw summary data file
raw_file_name = "Raw Data/FATE_DOC_202509.txt"

#file path and name for raw all peaks file
raw_allpeaks_name = "Raw Data/FATE_DOC_202509_allpeaks.txt"
```

```

#file path and name of processed data file
processed_file_name = "Processed Data/FATE_Processed_DOC_202509.csv"

##### Log Files - PLEASE CHECK
#downloaded metadata csv - downloaded from Google drive as csv for this year
#Raw_Metadata = "Raw Data/COMPASS_SynopticCB_PW_SampleLog_2025.csv"

#qaqc log file path for this year
Log_path = "Raw Data/FATE_TOCTN_QAQClog_2025.csv"

```

##Set Up Code

0.1 Import Data Functions

0.2 Import Sample Data

Import Sample Data

New names:

* ' -> '...14'

A tibble: 6 x 4

sample_name	npoc_raw	tdn_raw	run_datetime
<chr>	<dbl>	<dbl>	<chr>
1 FATE_Dock_20250327	5.04	0.0849	9/18/2025 10:57:13 AM
2 FATE_Flume_20250327	6.72	0.352	9/18/2025 11:31:08 AM
3 FATE_Dock_20250403	5.01	0.160	9/18/2025 12:04:30 PM
4 FATE_Flume_20250403	6.39	0.250	9/18/2025 12:36:56 PM
5 FATE_Dock_20250414	5.17	0.200	9/18/2025 1:09:31 PM
6 FATE_Flume_20250414	6.46	0.204	9/18/2025 1:41:44 PM

0.3 Assessing Standard Curves

Assess the Standard Curves

New names:

* ' -> '...18'

Warning: One or more parsing issues, call 'problems()' on your data frame for details,

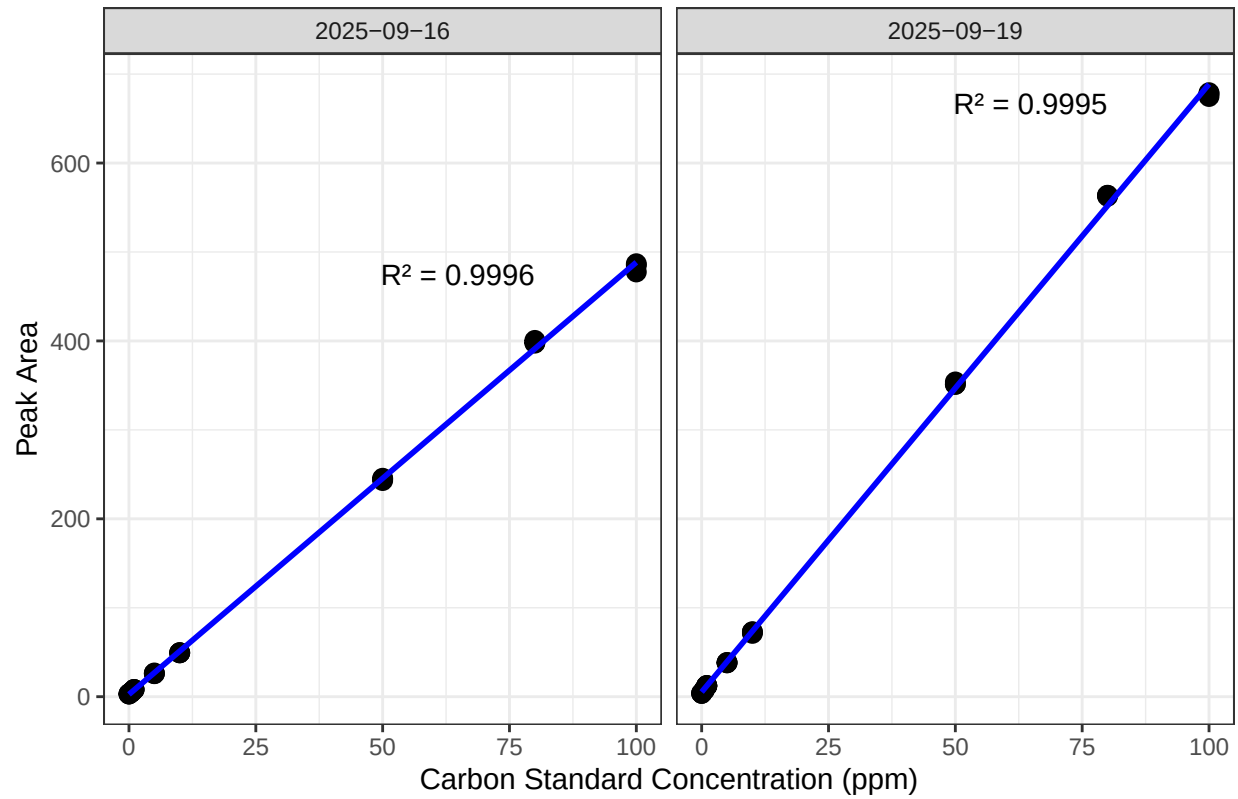
e.g.:

dat <- vroom(...)

problems(dat)

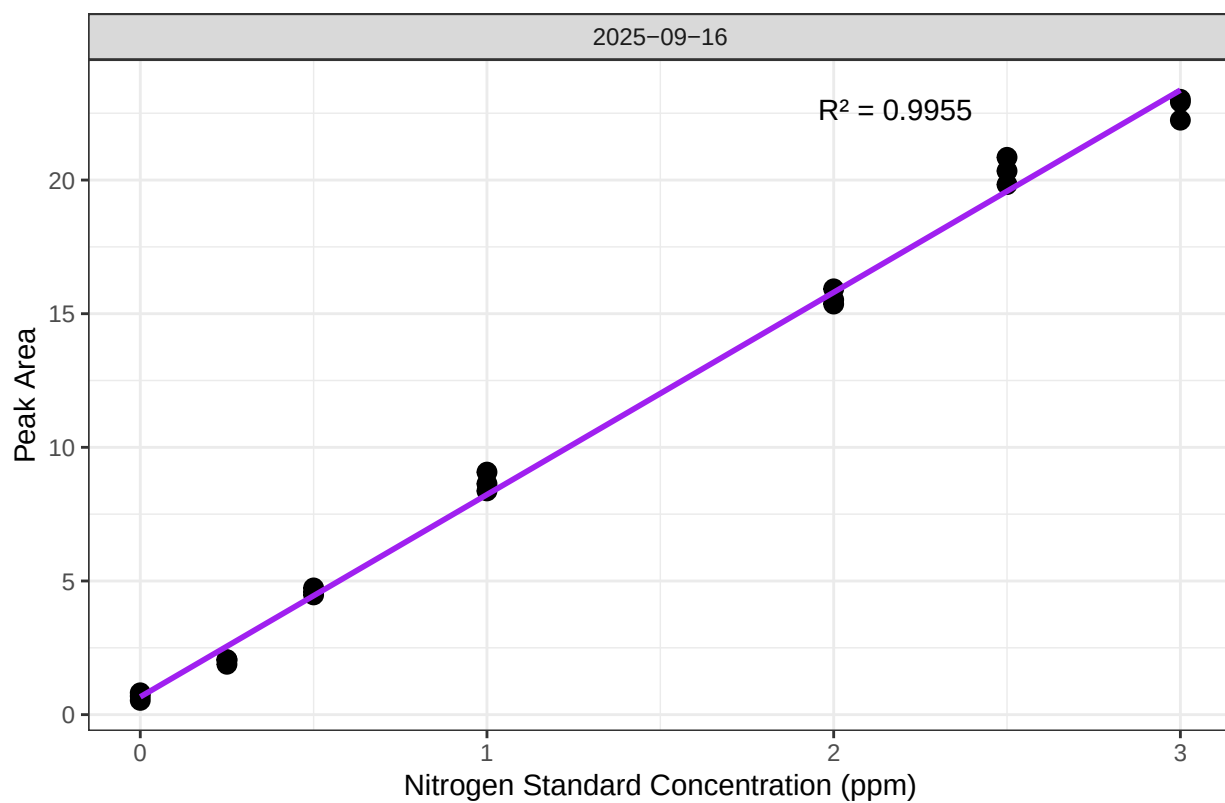
'geom_smooth()' using formula = 'y ~ x'

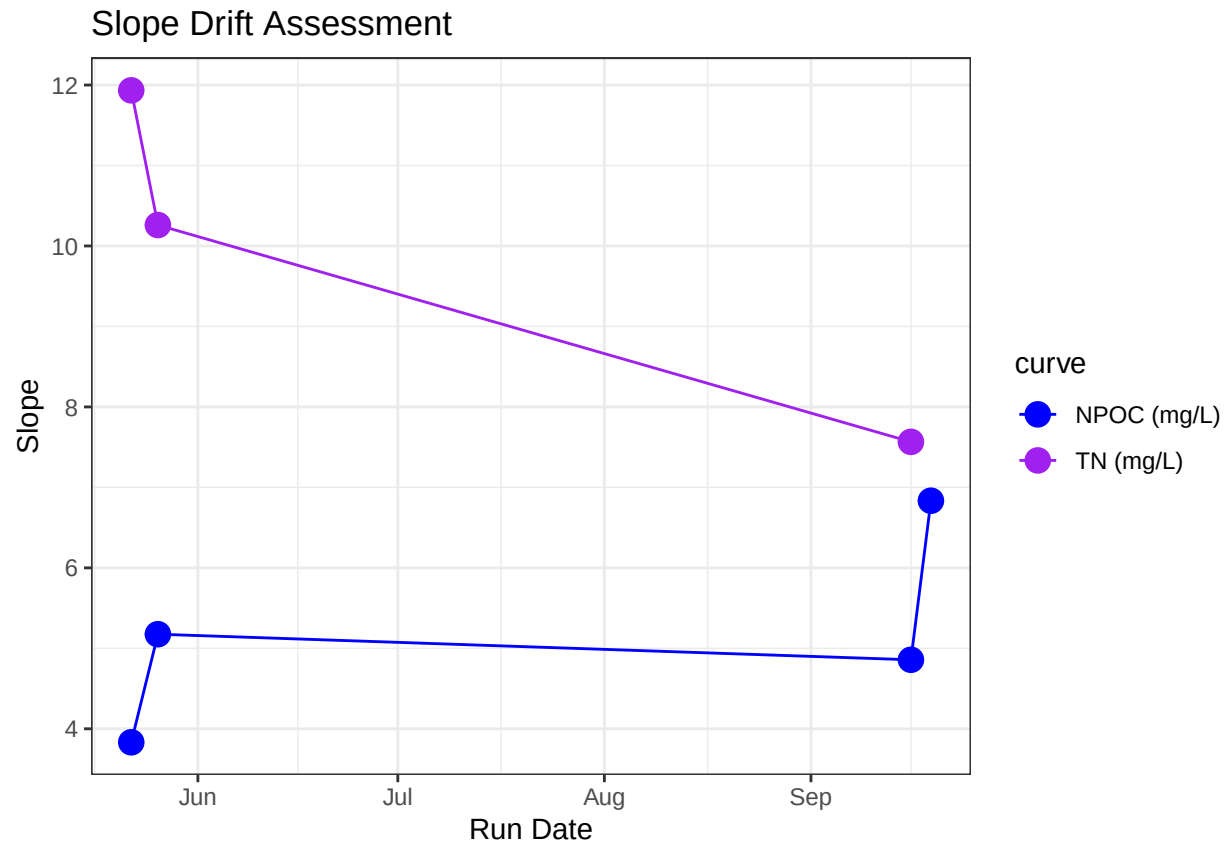
NPOC Std Curve by Date



```
## 'geom_smooth()' using formula = 'y ~ x'
```

TN Std Curve by Date





```
## [1] "NPOC Curve r2 GOOD"
```

```
## [1] "TN Curve r2 GOOD"
```

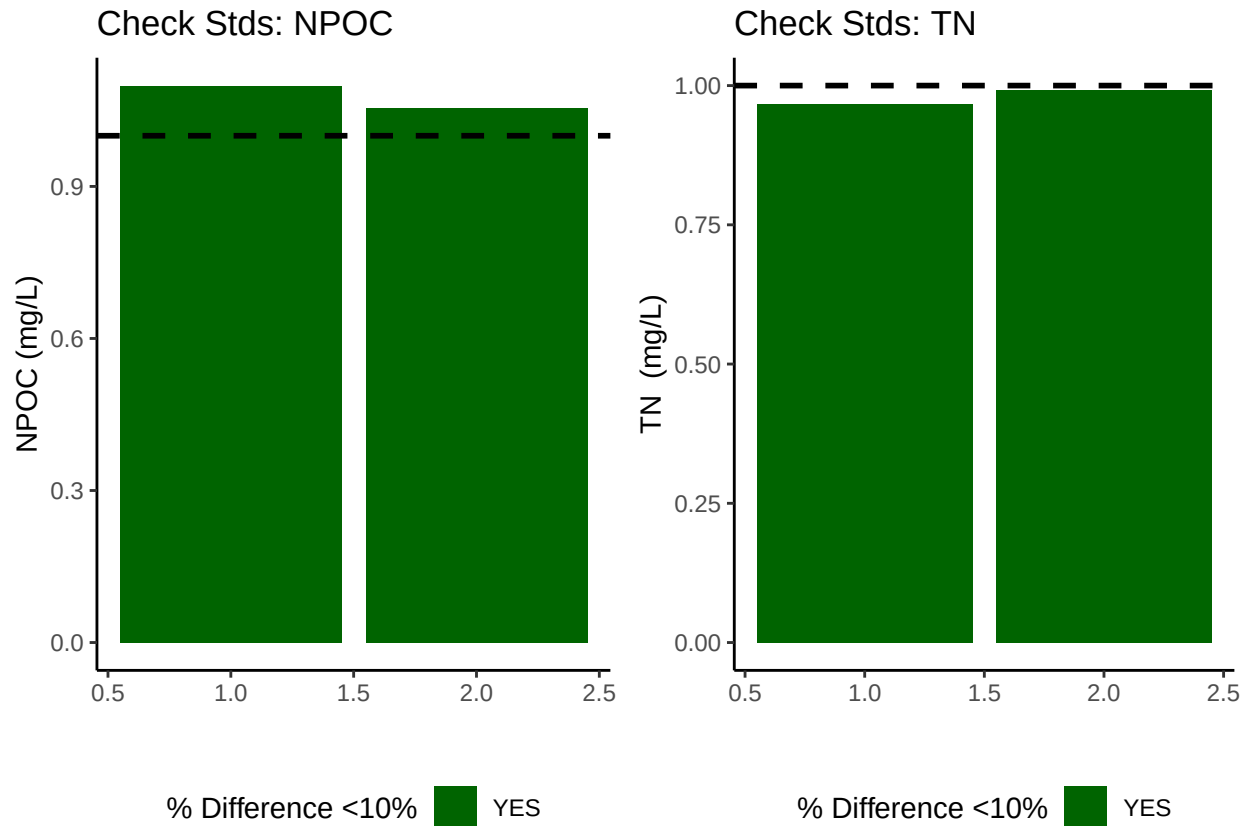
0.4 Assess Check Standards

```
## Assess the Check Standards
```

```
## New names:
## * ' ' -> '...14'
```

```
## [1] "Carbon Check Standard RSD within Range"
```

```
## [1] "Nitrogen Check Standard RSD within Range"
```



```
## [1] ">60% of Carbon Check Standards are within range of expected concentration"
```

```
## [1] ">60% of Nitrogen Check Standards are within range of expected concentration"
```

0.5 Assess Blanks

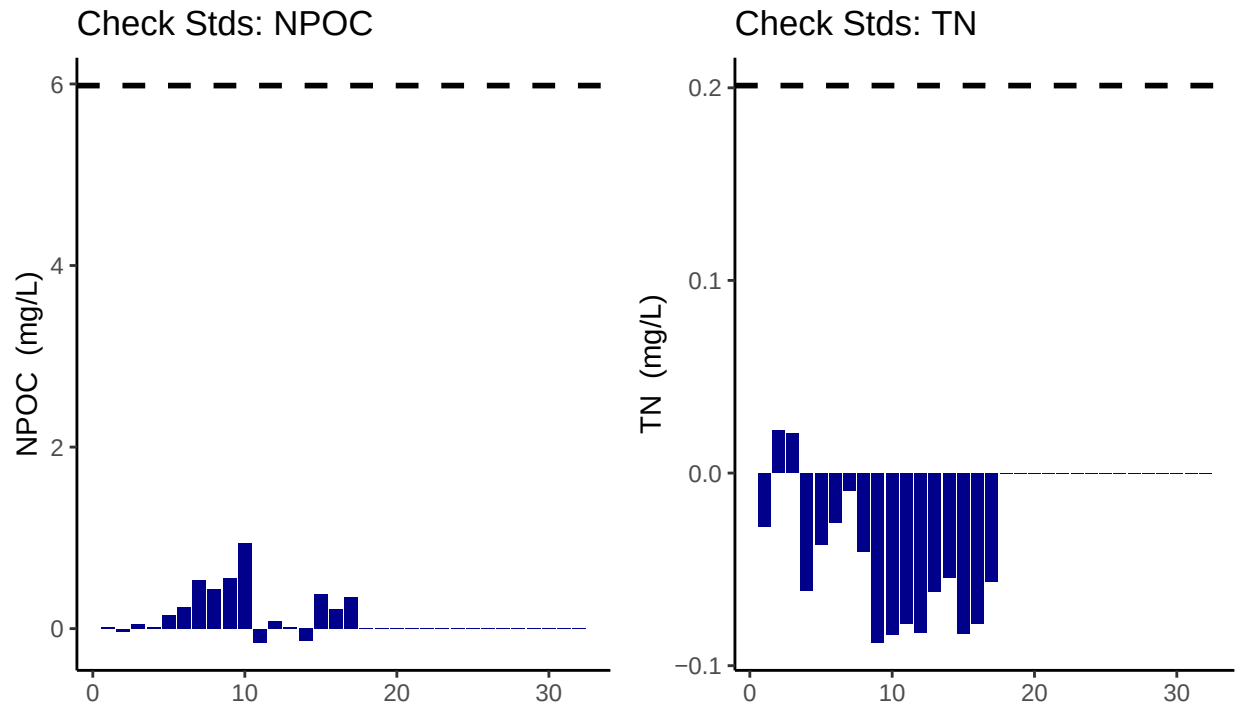
```
## Assess Blanks
```

```
## New names:
```

```
## * ' ' -> '...14'
```

```
## [1] ">60% of Carbon Blank concentrations are lower 25% quartile of samples"
```

```
## [1] ">60% of Nitrogen Blank concentrations are lower 25% quartile of samples"
```



Blank Conc <25% Quartile Samples ■ YE

Blank Conc <25% Quartile Samples ■ Y

```
## carbon blanks:
```

```
## [1] 0.1136484
```

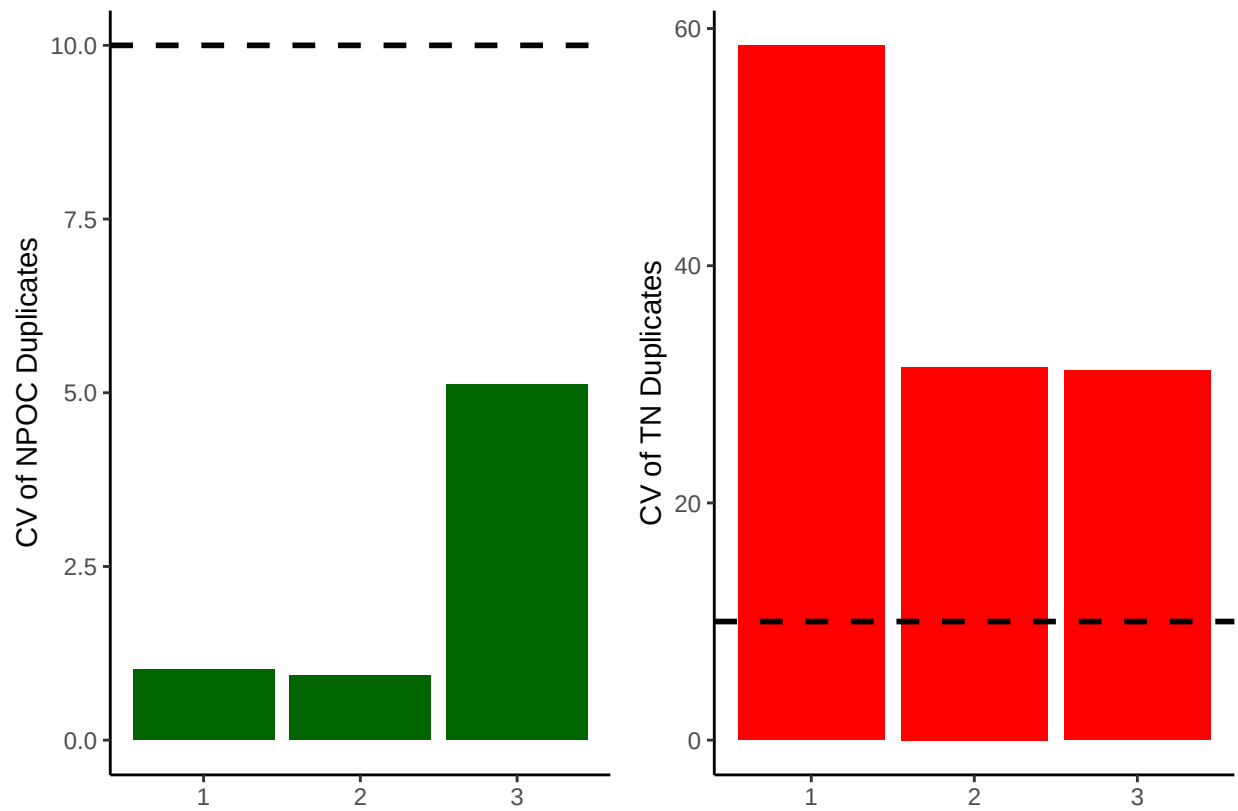
```
## nitrogen blanks:
```

```
## [1] -0.02578937
```

0.6 Assess Duplicates

```
## Assess Duplicates
```

```
## Warning: Using 'size' aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use 'linewidth' instead.
## This warning is displayed once every 8 hours.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.
```

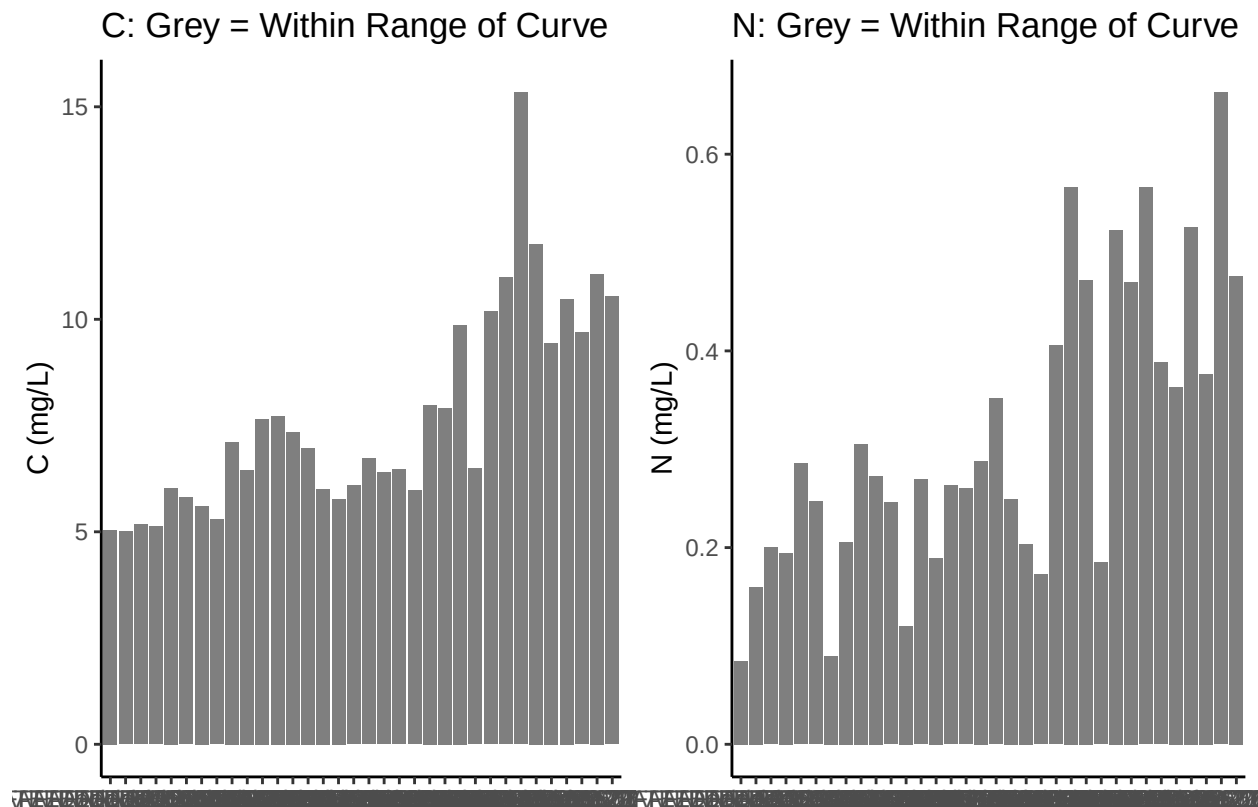


```
## [1] ">60% of Carbon Duplicates have a CV <10%"
```

```
## [1] "<60% of Nitrogen Duplicates have a CB <10% - REASSESS"
```

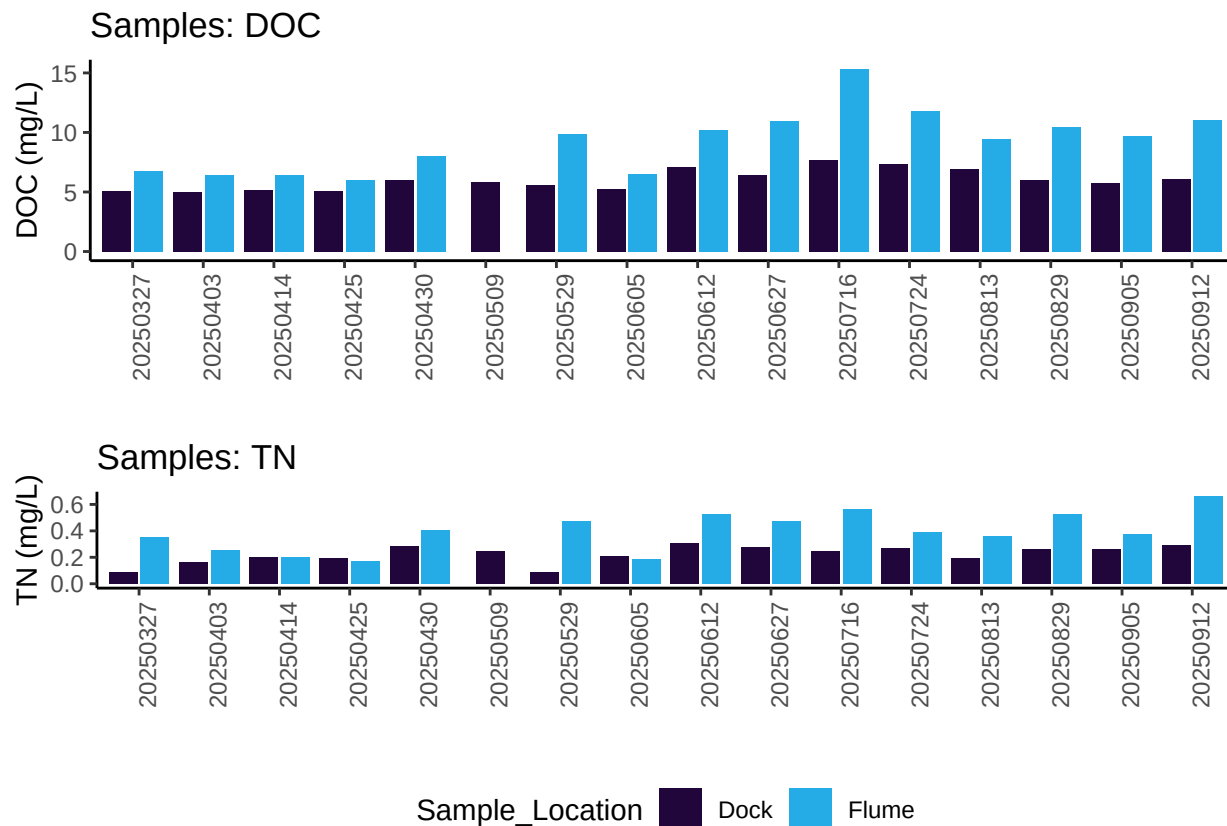
0.7 Sample Flagging - Are samples Within the range of the curve?

```
## Sample Flagging
```

0.8 Visualize Data by Plot

```
## Visualize Data
```



0.9 Convert data from mg/L to uMoles/L

0.10 Export Processed Data

Export Processed Data

```
## # A tibble: 6 x 11
##   Project Sample_Location Date      npoc_mgL npoc_uM npoc_flag tdn_mgL tdn_uM
##   <chr>    <chr>          <chr>      <dbl>    <dbl> <chr>      <dbl>    <dbl>
## 1 FATE     Dock             20250327    5.04    420. ""         0.0849    6.06
## 2 FATE     Flume            20250327    6.72    559. ""         0.352     25.1
## 3 FATE     Dock             20250403    5.01    417. ""         0.160     11.4
## 4 FATE     Flume            20250403    6.39    532. ""         0.250     17.8
## 5 FATE     Dock             20250414    5.17    431. ""         0.200     14.3
## 6 FATE     Flume            20250414    6.46    538. ""         0.204     14.5
## # i 3 more variables: tdn_flag <chr>, Analysis_runtime <chr>, Run_notes <chr>
```

#end